

Signorini's Problem

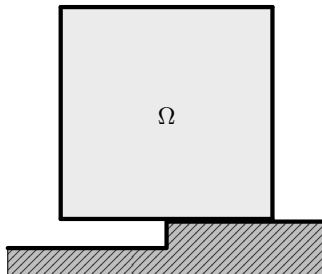
Max Stuthmann

March 18, 2026

- 1 Introduction – What's the Problem?
- 2 Geometrical and Physical Setup
 - Boundary Decomposition
 - Elasticity and Forces
 - Admissible Displacements
- 3 Energy Minimization
- 4 Weak Formulations
- 5 Strong Formulations
 - A Hybrid Formulation
 - Classical Strong Formulation

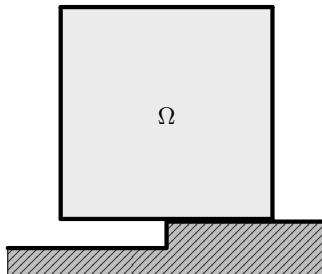
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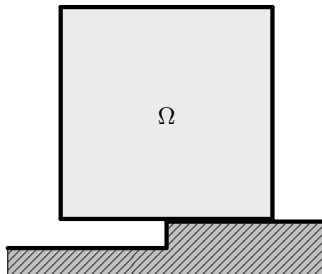
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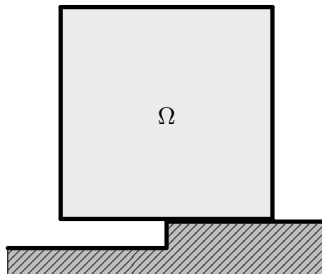
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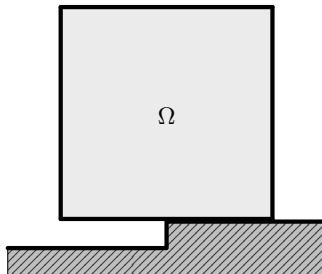
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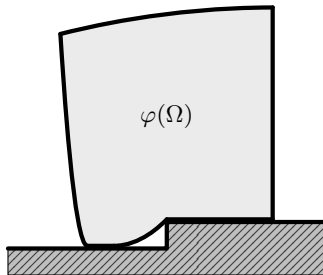
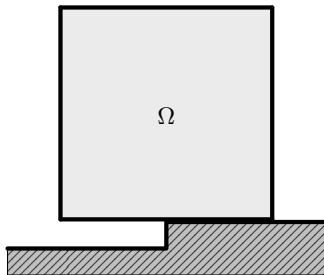
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- ▶ Ω may be subject to elasticity properties and internal/external forces.
- ▶ **Not in equilibrium configuration.**



What is to be found?

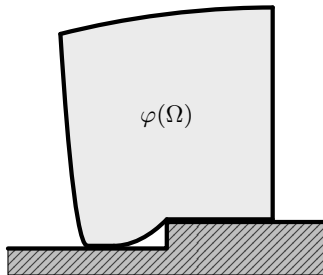
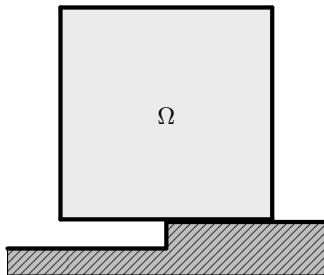
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$$\varphi(x) = x + u(x), \quad u : \bar{\Omega} \rightarrow \mathbb{R}^d.$$

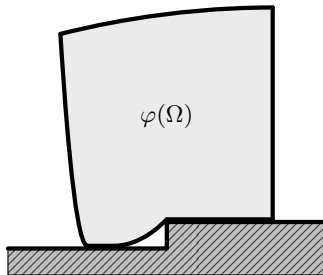
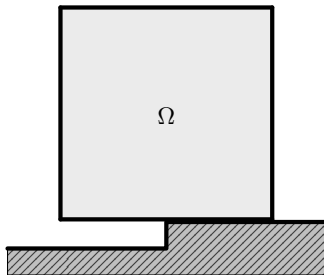


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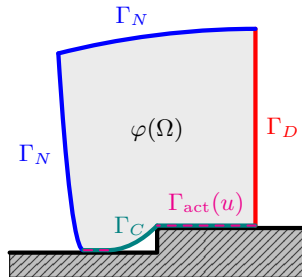
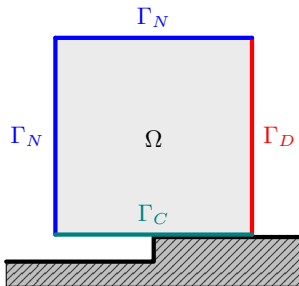
- ▶ We aim to find the *displacement function* u .



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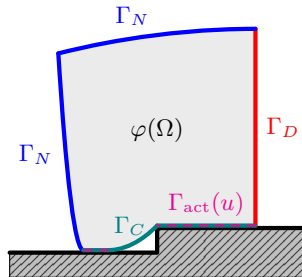
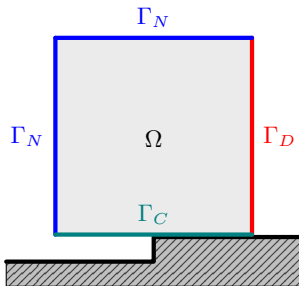
Boundary Decomposition and Preview on the Boundary Conditions

- Decompose $\Gamma = \partial\Omega$ into three relatively open and pairwise disjoint parts $\Gamma = \overline{\Gamma_D} \dot{\cup} \overline{\Gamma_N} \dot{\cup} \overline{\Gamma_C}$ with $|\Gamma_D|, |\Gamma_C| > 0$ and $|\Gamma_N| \geq 0$.



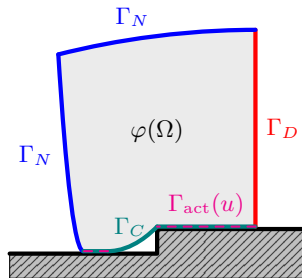
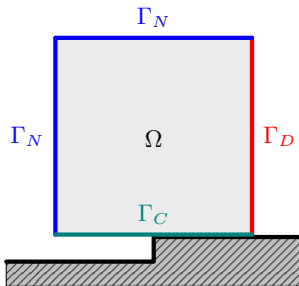
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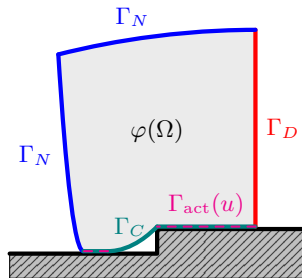
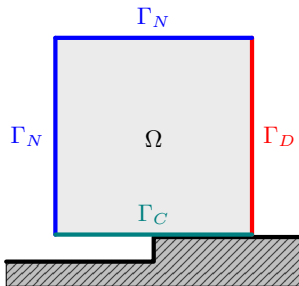
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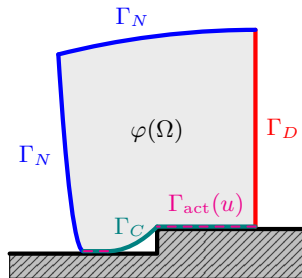
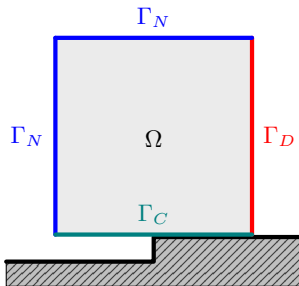
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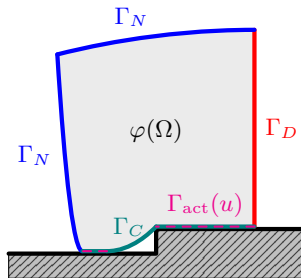
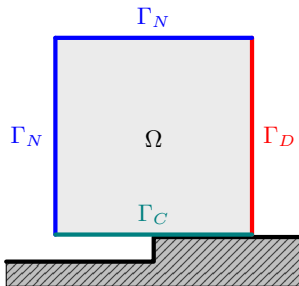
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- ▶ Rigid foundation can be describe by gap-function $g : \Gamma_C \rightarrow \mathbb{R}$. We assume $g = 0$.



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Basic Notations on Sobolev Spaces

- ▶ On $H^s(\Omega; \mathbb{R}^d)$ for $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in [0, 1)$, we use $(\cdot, \cdot)_{s, \Omega}$ and $\|\cdot\|_{s, \Omega}$

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- ▶ *Trace Theorem*: Restriction of smooth functions $\overline{\Omega} \rightarrow \mathbb{R}^d$ onto the boundary $\Gamma = \partial\Omega \in \mathcal{C}^{m,1}$ extends uniquely to a continuous surjective linear trace operator

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- ▶ For now, Ω is Lipschitz, i.e. $m = 0$ and we need $s = 1$, so

$$\gamma : H^1(\Omega; \mathbb{R}^d) \rightarrow H^{1/2}(\Gamma; \mathbb{R}^d)$$

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- If $\mathcal{A} : \Omega \rightarrow \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ is a fourth order tensor and $s \in \mathbb{S}^d$, we use the *double contraction*

$$s : \mathcal{A}(x) : s := \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- ▶ $\mathcal{A}_{ijkl} \in L^\infty(\Omega)$ is sufficient for $s = 1$. Later on, we need more.

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- ▶ *Work = force · displacement:* The linear functional

$$\ell(v) := \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad v \in H^1(\Omega; \mathbb{R}^d)$$

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- ▶ *Equilibrium* between both: $a(u, v) = \ell(v) \quad \forall v \in H^1(\Omega; \mathbb{R}^d)$.

Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : V \times V \rightarrow \mathbb{R}$ defined above is

- i) *bounded, i.e. for all $u, v \in V$ there exists $C_1 < \infty$ such that*

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega}.$$

- ii) *V -elliptic, i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2.$$

Moreover, the linear functional ℓ defined above is bounded, i.e. for all $v \in V$ there exists $C_3 > 0$ such that

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega}.$$

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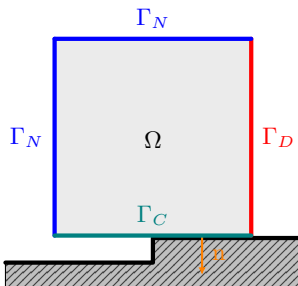
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Admissible Displacements and the Nonpenetration Constraint

- *Homogeneous Dirichlet* boundary conditions on Γ_D are incorporated in the underlying Hilbert space

$$V := \{v \in H^1(\Omega; \mathbb{R}^d) \mid (\gamma v)|_{\Gamma_D} = 0\}$$

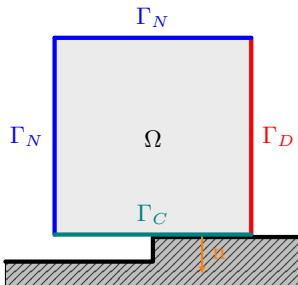


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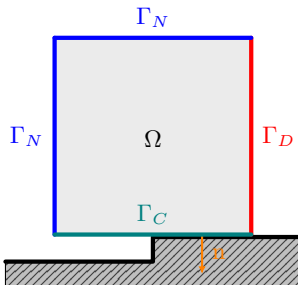


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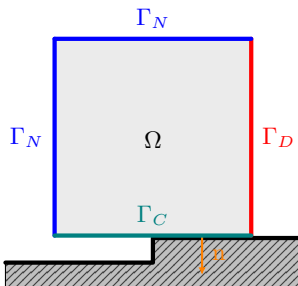


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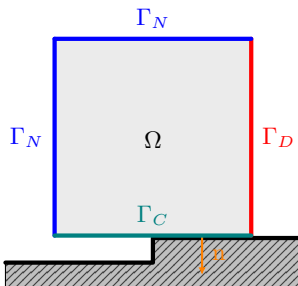


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- ▶ K is a closed and convex cone in V , i.e. if $v \in K$ then $\lambda v \in K$ for all $\lambda \in [0, \infty)$.



► *Equilibrium:*

$$a(u, v) = \int_{\Omega} \sigma(u) : \varepsilon(v) \, dx = \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} F \cdot (\gamma v) \, ds = \ell(v) \quad \forall v \in V$$

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Problem (Signorini's Problem – Energy Minimization)

$$\text{Find } u \in K \text{ such that: } \mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v).$$

Proposition (Well-Posedness of Signorini's Problem)

There exists exactly one $u \in K$ that minimizes $\mathcal{J}$ over K . Moreover, the unique minimizer satisfies the a-priori bound

$$\|u\|_{1,\Omega} \leq \frac{2C_3}{C_2} \|\ell\|_{V'}.$$

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- ▶ Stability bound follows from V -ellipticity of a w.r.t. C_2 and boundedness of ℓ w.r.t. C_3 .

Proposition (First-order optimality condition on convex sets)

A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

$$\mathcal{J}'(u)[v - u] = a(u, v - u) - \ell(v - u) \geq 0 \quad \forall v \in K.$$

Weak Formulations / Variational Inequality

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Problem (Signorini's Problem – Weak Formulation 2)

Find $u \in K$ such that:
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Content

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- 2 Geometrical and Physical Setup
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- 4 Weak Formulations
- 5 Strong Formulations
 - A Hybrid Formulation
 - Classical Strong Formulation

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- ▶ Then $\operatorname{div} \sigma(u) \in L^2(\Omega; \mathbb{R}^d)$ and we can understand normal traces on $H(\operatorname{div}; \Omega)$ in the following way.

Normal Traces on $H(\operatorname{div}; \Omega)$

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$$\langle \gamma_n(\tau), \gamma v \rangle_\Gamma := F_\tau(v), \quad \gamma_n(\tau) \in H^{-1/2}(\Gamma; \mathbb{R}^d).$$

- ▶ In the smooth case, $\gamma_n(\tau)$ coincides with the classical traction $(\gamma\tau)_n$.

Stress Regularity Assumption

- ▶ Thus, we further assume $\sigma(u) \in H(\operatorname{div}; \Omega)$ which yields $\sigma(u)_n := \gamma_n(\sigma(u)) \in H^{-1/2}(\Gamma; \mathbb{R}^d)$.

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- ▶ *Useful identity:* If $-\operatorname{div} \sigma(u) = f$ almost everywhere in Ω , then

$$a(u, v) - \ell(v) = \langle \sigma(u)n, \gamma v \rangle_\Gamma - \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad \forall v \in V.$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$, find $u \in K$ such that $\sigma(u) \in H(\text{div}; \Omega)$ and

$$\begin{cases} -\text{div } \sigma(u) = f & \text{a.e. in } \Omega, \end{cases} \quad (1a)$$

$$\begin{cases} \langle \sigma(u)n, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot (\gamma u) \, ds, \end{cases} \quad (1b)$$

$$\begin{cases} \langle \sigma(u)n, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v) \, ds \quad \forall v \in K. \end{cases} \quad (1c)$$

Equivalence of Hybrid and Weak Formulation

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\operatorname{div}; \Omega)$. Then $u \in K$ solves the hybrid problem above if and only if it solves the weak problem

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- ▶ Subtract (1b) from (1c):

$$\langle \sigma(u)n, \gamma(v - u) \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma(v - u) \, ds \geq 0 \quad \forall v \in K.$$

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$$\langle \sigma(u)n, \gamma(v - u) \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma(v - u) \, ds \geq 0 \quad \forall v \in K.$$

- ▶ Insert the identity from the above:

$$a(u, v - u) - \ell(v - u) \geq 0 \quad \forall v \in K,$$

i.e. the weak formulation.

Equivalence of Hybrid and Weak Formulation

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\operatorname{div}; \Omega)$. Then $u \in K$ solves the hybrid problem above if and only if it solves the weak problem

$$\text{Find } u \in K \text{ such that} \quad a(u, v - u) \geq \ell(v - u) \quad \forall v \in K.$$

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► Let $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$. Then $u \pm t\varphi \in K$ for $t \in \mathbb{R}$ because $\gamma\varphi = 0$, hence the weak inequality implies

$$a(u, \varphi) = \ell(\varphi) = \int_{\Omega} f \cdot \varphi \, dx = \langle f, \varphi \rangle \quad \forall \varphi \in \mathcal{D}(\Omega; \mathbb{R}^d).$$

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- ▶ Thus, $-\operatorname{div} \sigma(u) = f$ in $\mathcal{D}'(\Omega; \mathbb{R}^d)$ and by $\sigma(u) \in H(\operatorname{div}; \Omega)$, it holds in $L^2(\Omega; \mathbb{R}^d)$ as well.

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- ▶ This is exactly (1b).

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► With the identity earlier, and the variational inequality, we have for all $v \in V$ that

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- ▶ This means

$$\langle \sigma(u)n, \gamma v \rangle_{\Gamma} \geq \int_{\Gamma_N} F \cdot \gamma v \, ds + \underbrace{\left[\langle \sigma(u)n, \gamma u \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma u \, ds \right]}_{=0} \quad \forall v \in V.$$

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- ▶ Normal Trace: $t(u) := (\gamma\sigma(u))n \in H^{s-3/2}(\Gamma; \mathbb{R}^d)$ is well-defined.
- ▶ If $\sigma(u) \in H(\text{div}; \Omega)$, the two concepts of normal traces coincide:

$$\langle \sigma(u)n, \gamma v \rangle_\Gamma = \int_{\Gamma_N} t(u) \cdot (\gamma v) \, ds \quad \forall v \in V.$$

Normal and Tangential Components

- Normal and tangential splitting of $t(u)$ into

$$\sigma_n(u) := t(u) \cdot \mathbf{n} \in H^{s-3/2}(\Gamma), \quad \sigma_t(u) := t(u) - \sigma_n(u) \mathbf{n} \in H^{s-3/2}(\Gamma; \mathbb{R}^d).$$

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- Furthermore, for $u \in H^s(\Omega; \mathbb{R}^d)$ we have $\gamma u \in (H^{s-1/2}(\Gamma; \mathbb{R}^d))$ with normal/tangential splitting

$$u_n := (\gamma u) \cdot \mathbf{n} \in H^{s-1/2}(\Gamma) \quad \text{and} \quad u_t := \gamma u - u_n \mathbf{n} \in H^{s-1/2}(\Gamma; \mathbb{R}^d)$$

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Classical Strong Formulation under Additional Regularity

Problem (Signorini's Problem – Classical Strong Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$ find $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ such that the small strain elasticity equations

$$\begin{cases} -\operatorname{div} \sigma(u) = f & \text{in } \Omega, \end{cases} \quad (2a)$$

$$\begin{cases} u = 0 & \text{on } \Gamma_D, \end{cases} \quad (2b)$$

$$\begin{cases} t(u) = F & \text{on } \Gamma_N, \end{cases} \quad (2c)$$

and the contact conditions

$$\begin{cases} u_n \leq 0 & \text{on } \Gamma_C, \end{cases} \quad (3a)$$

$$\begin{cases} \sigma_n(u) \leq 0 & \text{on } \Gamma_C, \end{cases} \quad (3b)$$

$$\begin{cases} \sigma_n(u) u_n = 0 & \text{on } \Gamma_C, \end{cases} \quad (3c)$$

$$\begin{cases} \sigma_t(u) = 0 & \text{on } \Gamma_C \end{cases} \quad (3d)$$

hold.

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- ▶ *Equilibrium equation* – $\operatorname{div} \sigma(u) = f$ in Ω : balance of internal stresses and body forces in the bulk.

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- ▶ *Frictionless contact* $\sigma_t(u) = 0$ on Γ_C : Foundation does not exert any tangential force on the body.

Proposition (Equivalence of Strong and Hybrid Formulation)

Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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- ▶ $u = 0$ on Γ_D (2b) and $u_n \leq 0$ on Γ_C (3a) are incorporated in the requirement $u \in K$.

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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► With $t(u) = \sigma_n(u)n + \sigma_t(u)$ and $\gamma v = v_n n + v_t$, we have

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- ▶ It remains to prove $t(u) = F$ on Γ_N and the remaining contact conditions.

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- ▶ Plug $\pm v$ into hybrid inequality:

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- ▶ Therefore,

$$\int_{\Gamma_N} (t(u) - F) \cdot g \, ds = 0 \quad \forall g \in H_{00}^{1/2}(\Gamma_N; \mathbb{R}^d) \quad \implies \quad t(u) = F \text{ on } \Gamma_N.$$

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